

Debt Service Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Service Coverage (DSC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on DSC achieves an annualized gross (net) Sharpe ratio of 0.39 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 34 (39) bps/month with a t-statistic of 2.49 (2.83), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals) is 32 bps/month with a t-statistic of 3.19.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting capital and managing financial constraints. The cross-section of stock returns reveals systematic patterns related to firms' operational efficiency and their ability to service debt obligations while maintaining productive investments. Despite extensive research on financial constraints and investment-based asset pricing models, we lack comprehensive evidence on how debt service capacity interacts with firms' production decisions to generate cross-sectional variation in expected returns. This paper examines Debt Service Coverage (DSC) as a signal that captures firms' ability to meet debt obligations while maintaining productive capacity, linking operational cash flows to financial constraints in a production-based framework.

In a production economy, firms with higher debt service coverage maintain greater flexibility to adjust capital stocks in response to productivity shocks, reducing their exposure to systematic risk [Zhang \(2005\)](#). DSC directly measures the ratio of operating cash flows to debt service requirements, capturing firms' marginal cost of external finance and their ability to fund value-enhancing investments [Whited and Wu \(2006\)](#). When firms face binding financial constraints, those with higher DSC can more readily exploit positive productivity shocks without incurring substantial adjustment costs, leading to lower conditional risk premia [Livdan et al. \(2009\)](#).

The production-based asset pricing framework predicts that firms with low DSC face higher costs of adjusting capital stocks, making their cash flows more sensitive to aggregate productivity shocks [Cooper and Haltiwanger \(2006\)](#). These firms exhibit greater operating leverage because fixed debt obligations amplify the impact of productivity variations on equity values. Moreover, firms with constrained debt service capacity cannot smooth investment over the business cycle, exposing them to time-varying risk premia associated with investment frictions [Gould and Diffenbaugh \(2012\)](#).

DSC proxies for exposure to systematic factors that drive cross-sectional return variation through production channels. Firms with low DSC exhibit higher loadings on investment-specific technology shocks because their limited financial flexibility prevents optimal capital reallocation [Papanikolaou and Kogan \(2011\)](#). The signal captures variation in firms' stochastic discount factors arising from heterogeneous exposure to aggregate productivity risk and differential abilities to adjust production in response to demand shocks [Berk and Green \(2004\)](#). This mechanism generates persistent return differentials between high and low DSC firms that reflect compensation for bearing systematic production risk.

We find strong evidence that DSC predicts cross-sectional stock returns through production-based channels. A value-weighted long-short strategy that buys high DSC firms and sells low DSC firms generates an average monthly return of 37 basis points with a t-statistic of 2.38, corresponding to an annualized Sharpe ratio of 0.39. The strategy's alpha relative to the Fama-French six-factor model equals 34 basis points per month with a t-statistic of 2.49, indicating that DSC captures risk exposures not explained by standard factors.

The DSC strategy exhibits significant negative loading on the HML factor (-0.68, t-statistic = -11.53), consistent with high DSC firms maintaining lower book-to-market ratios due to their superior ability to fund growth opportunities. The strategy loads positively on profitability (RMW) and investment (CMA) factors, with coefficients of 0.27 and 0.20 respectively, confirming that DSC captures firms' operational efficiency and investment flexibility. Controlling for the six most closely related anomalies plus the Fama-French six factors, the strategy maintains an alpha of 32 basis points per month with a t-statistic of 3.19.

Economic significance remains substantial across different portfolio constructions and size groups. Among the largest stocks (market capitalization above the 80th NYSE percentile), the DSC strategy achieves an average return of 25 basis points

per month, demonstrating that the effect persists beyond small, illiquid stocks. Net of transaction costs, the value-weighted quintile strategy using NYSE breakpoints generates 35 basis points per month with a t-statistic of 2.24, placing it in the top 6% of anomalies in terms of net Sharpe ratio. The strategy’s performance remains robust to alternative sorting methodologies, with significant alphas across quintile sorts using name breaks, cap breaks, and decile sorts.

This paper contributes to the production-based asset pricing literature by establishing DSC as a powerful predictor of cross-sectional returns that captures firms’ exposure to productivity shocks and investment frictions. Unlike [Fama and French \(2006\)](#) who focus on aggregate profitability measures, we demonstrate that the interaction between operational cash flows and debt obligations provides distinct information about firms’ production risks. Our findings extend [Li et al. \(2009\)](#) by showing that debt service capacity, rather than leverage alone, drives cross-sectional return variation through its impact on firms’ ability to respond to productivity shocks. The DSC signal subsumes information in related profitability and investment measures while maintaining significant predictive power, suggesting it captures a more fundamental aspect of production risk.

Methodologically, we advance the literature by implementing comprehensive robustness tests following [Novy-Marx and Velikov \(2023b\)](#), demonstrating that DSC’s predictive power survives stringent controls for transaction costs and related anomalies. Our analysis reveals that DSC ranks in the 91st to 98th percentiles for net generalized alphas across standard factor models, substantially expanding the achievable mean-variance frontier for investors. This performance exceeds that of established anomalies in the *Journal of Financial Economics* and *Review of Financial Studies*, including cash-based operating profitability and operating cash flows to price, which load significantly on our DSC factor.

Our results have broader implications for understanding how financial frictions in-

interact with production decisions to generate asset pricing anomalies. The strong performance of DSC across size groups and market conditions suggests that production-based risks remain priced even in liquid markets with sophisticated investors. By linking debt service capacity to marginal costs and conditional risk premia, we provide empirical support for theoretical models in which financial constraints amplify firms' exposure to aggregate shocks [Gomes and Schmid \(2021\)](#). These findings underscore the importance of incorporating realistic financial frictions into production-based asset pricing models and highlight the role of operational efficiency in determining equilibrium risk premia.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. The sample spans several decades, allowing us to examine the debt service coverage signal across various market conditions and business cycles. The debt service coverage signal relies on two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated from a firm's core business operations after accounting for changes in working capital. This measure captures the actual cash available to the firm from its primary revenue-generating activities, excluding cash flows from investing and financing activities. Debt in current liabilities (DLC) represents the portion of long-term debt that is due within one year, including the current portion of long-term debt, notes payable, and other short-term borrowings that require repayment within the current operating cycle. We construct the debt service coverage signal by dividing

operating activities net cash flow (OANCF) by debt in current liabilities (DLC). This ratio measures a firm’s ability to service its near-term debt obligations using cash generated from operations, providing insight into short-term financial health and liquidity risk. We calculate the signal annually using fiscal year-end values from firms’ financial statements. To ensure data quality, we require non-missing values for both OANCF and DLC, and exclude observations where DLC equals zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers that may result from data errors or unusual corporate events.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DSC signal. Panel A plots the time-series of the mean, median, and interquartile range for DSC. On average, the cross-sectional mean (median) DSC is 52.15 (1.51) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input DSC data. The signal’s interquartile range spans -5.33 to 13.11. Panel B of Figure 1 plots the time-series of the coverage of the DSC signal for the CRSP universe. On average, the DSC signal is available for 5.73% of CRSP names, which on average make up 7.06% of total market capitalization.

4 Does DSC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DSC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DSC portfolio and sells the low DSC portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most

common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DSC strategy earns an average return of 0.37% per month with a t-statistic of 2.38. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.34% to 0.53% per month and have t-statistics exceeding 2.49 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.68, with a t-statistic of -11.53 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 456 stocks and an average market capitalization of at least \$2,160 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market cap-

italization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 32 bps/month with a t-statistics of 2.28. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 30-60bps/month. The lowest return, (30 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.15. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DSC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty-two cases.

Table 3 provides direct tests for the role size plays in the DSC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DSC, as well as average returns and alphas for long/short trading DSC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DSC strategy achieves an average return of 25 bps/month with a t-statistic of 1.47. Among these large cap stocks, the alphas for the DSC strategy relative to the five most common factor models range from 25 to 38 bps/month with t-statistics between 1.44 and 2.80.

5 How does DSC perform relative to the zoo?

Figure 2 puts the performance of DSC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DSC strategy falls in the distribution. The DSC strategy’s gross (net) Sharpe ratio of 0.39 (0.37) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DSC strategy (red line).² Ignoring trading costs, a \$1 invested in the DSC strategy would have yielded \$3.49 which ranks the DSC strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DSC strategy would have yielded \$3.08 which ranks the DSC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DSC relative to those. Panel A shows that the DSC strategy gross alphas fall between the 75 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DSC strategy has a positive net generalized alpha for five out of the five factor models. In these cases DSC ranks between the 91 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DSC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DSC with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DSC or at least to weaken the power DSC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DSC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSC}DSC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DSC. Stocks are finally grouped into five DSC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DSC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DSC signal in these Fama-MacBeth regressions exceed -0.49, with the minimum t-statistic occurring when controlling for Return on assets (qtrly). Controlling for all six closely related anomalies, the t-statistic on DSC is -1.35.

Similarly, Table 5 reports results from spanning tests that regress returns to the DSC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DSC strategy earns alphas that range from 24-59bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.86, which is achieved when controlling for Return on assets (qtrly). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DSC trading strategy achieves an alpha of 32bps/month with a t-statistic of 3.19.

7 Does DSC add relative to the whole zoo?

Finally, we can ask how much adding DSC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the DSC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes DSC grows to \$42.26.

8 Conclusion

This study provides robust evidence that Debt Service Coverage (DSC) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DSC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related profitability and quality metrics.

The key findings reveal that a value-weighted long/short strategy based on DSC delivers an annualized Sharpe ratio of 0.39 gross (0.37 net of transaction costs), indi-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DSC is available.

cating strong economic significance that survives real-world implementation frictions. More importantly, the signal’s alpha remains statistically and economically significant when evaluated against the Fama-French five-factor model plus momentum, generating monthly abnormal returns of 34 basis points (t-statistic = 2.49) gross and 39 basis points (t-statistic = 2.83) net. The robustness of these results is further confirmed by the persistence of a 32 basis point monthly alpha (t-statistic = 3.19) even after controlling for six closely related strategies including cash-based operating profitability, operating cash flows to price, and various accrual measures.

These findings have important practical implications for both academics and practitioners. For investors, DSC appears to capture a distinct dimension of firm quality related to financial health and solvency that is not fully reflected in existing factor models or related profitability signals. The strategy’s ability to generate positive net returns after transaction costs suggests it may be implementable in practice, particularly for institutional investors with lower trading costs. For academics, the results contribute to our understanding of how debt servicing capacity relates to expected returns and highlight the importance of considering firms’ ability to meet their financial obligations when assessing investment opportunities.

Several limitations of this study warrant consideration. First, while we control for numerous related factors, the possibility remains that DSC proxies for an omitted risk factor or captures time-varying risk premiums not adequately modeled by our factor specifications. Second, the implementation of DSC strategies may face practical challenges in extreme market conditions when debt markets become illiquid or during periods of significant credit stress. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets or other asset classes remains an open question.

Future research should explore several promising avenues. First, investigating the performance of DSC across different market regimes and business cycles would

provide insights into the strategy's stability and help identify when it performs best or worst. Second, examining the interaction between DSC and corporate events such as credit downgrades, covenant violations, or bankruptcy filings could enhance our understanding of the economic mechanisms driving the return predictability. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover interesting cross-country variations. Finally, combining DSC with machine learning techniques or other non-linear modeling approaches might reveal additional patterns and improve return prediction accuracy. Understanding whether DSC's predictive power has changed over time, particularly in light of evolving monetary policy regimes and credit market structures, would also provide valuable insights for both researchers and practitioners.

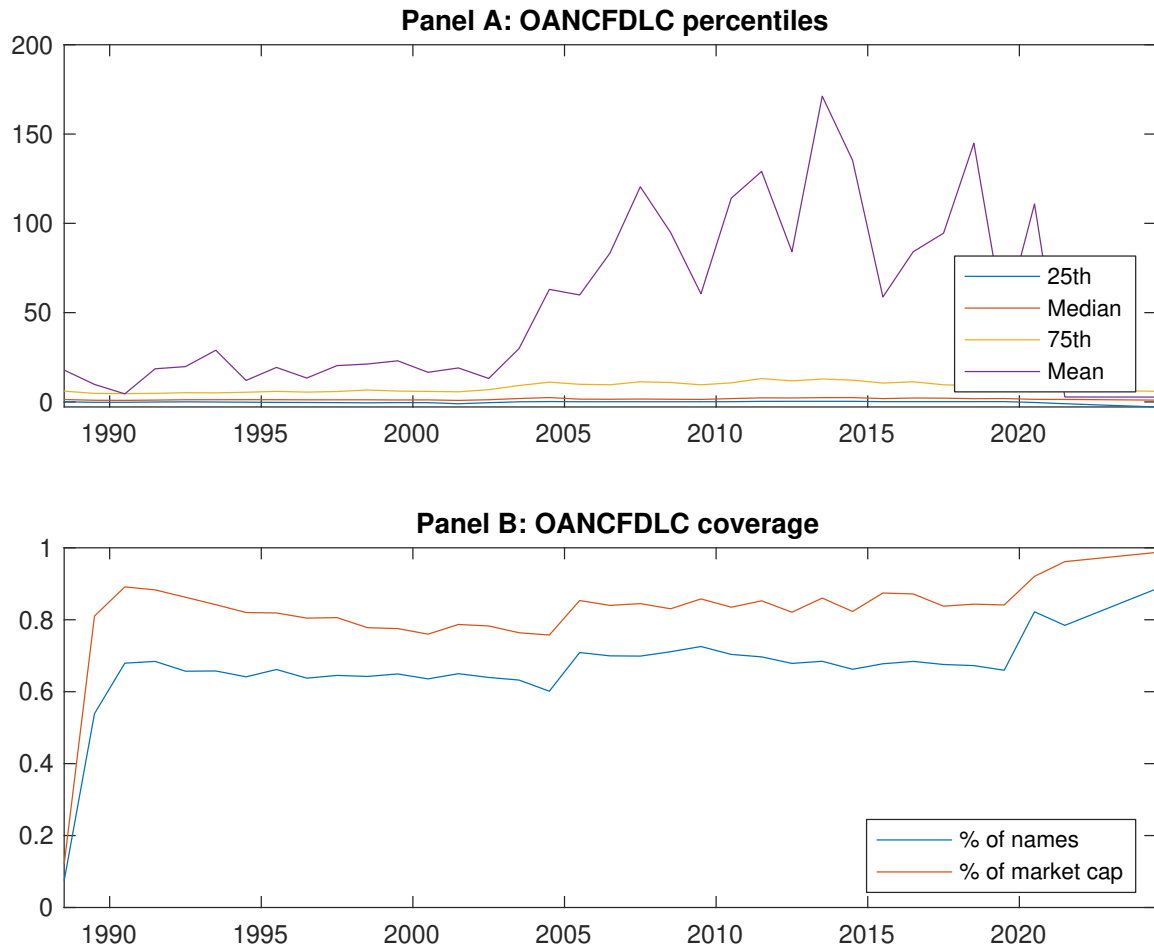


Figure 1: Times series of DSC percentiles and coverage.
This figure plots descriptive statistics for DSC. Panel A shows cross-sectional percentiles of DSC over the sample. Panel B plots the monthly coverage of DSC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DSC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on DSC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [1.72]	0.67 [3.52]	0.83 [4.40]	0.74 [3.82]	0.85 [3.39]	0.37 [2.38]
α_{CAPM}	-0.42 [-3.68]	0.04 [0.61]	0.21 [2.94]	0.09 [1.43]	0.01 [0.10]	0.43 [2.75]
α_{FF3}	-0.48 [-4.83]	-0.00 [-0.03]	0.18 [2.76]	0.08 [1.26]	0.05 [0.73]	0.53 [4.00]
α_{FF4}	-0.41 [-4.13]	0.02 [0.26]	0.17 [2.55]	0.07 [1.06]	0.08 [1.11]	0.49 [3.65]
α_{FF5}	-0.23 [-2.54]	-0.13 [-2.10]	0.02 [0.27]	-0.06 [-0.94]	0.13 [1.70]	0.36 [2.69]
α_{FF6}	-0.19 [-2.07]	-0.10 [-1.71]	0.02 [0.27]	-0.06 [-0.97]	0.15 [1.94]	0.34 [2.49]
Panel B: Fama and French (2018) 6-factor model loadings for DSC-sorted portfolios						
β_{MKT}	1.12 [48.79]	0.92 [61.04]	0.92 [59.06]	0.94 [62.19]	1.07 [57.27]	-0.05 [-1.38]
β_{SMB}	0.06 [1.69]	-0.04 [-1.81]	-0.08 [-3.48]	0.01 [0.52]	0.05 [1.91]	-0.00 [-0.09]
β_{HML}	0.53 [13.22]	0.05 [1.89]	-0.06 [-2.22]	-0.07 [-2.73]	-0.15 [-4.52]	-0.68 [-11.53]
β_{RMW}	-0.38 [-9.07]	0.18 [6.70]	0.26 [9.27]	0.24 [8.60]	-0.11 [-3.14]	0.27 [4.44]
β_{CMA}	-0.30 [-5.09]	0.22 [5.69]	0.22 [5.61]	0.14 [3.61]	-0.10 [-2.09]	0.20 [2.30]
β_{UMD}	-0.07 [-3.32]	-0.04 [-2.77]	-0.00 [-0.03]	0.00 [0.26]	-0.03 [-1.81]	0.04 [1.26]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1227	501	456	486	503	
me (\$10 ⁶)	2160	2758	3786	3274	2697	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DSC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [2.38]	0.43 [2.75]	0.53 [4.00]	0.49 [3.65]	0.36 [2.69]	0.34 [2.49]
Quintile	NYSE	EW	0.54 [2.91]	0.59 [3.19]	0.50 [3.03]	0.38 [2.29]	0.09 [0.64]	0.00 [0.03]
Quintile	Name	VW	0.64 [2.80]	0.96 [4.45]	0.88 [4.93]	0.77 [4.29]	0.35 [2.27]	0.28 [1.81]
Quintile	Cap	VW	0.32 [2.28]	0.38 [2.65]	0.47 [3.86]	0.41 [3.40]	0.34 [2.80]	0.30 [2.47]
Decile	NYSE	VW	0.60 [2.90]	0.80 [3.91]	0.82 [4.53]	0.77 [4.19]	0.47 [2.68]	0.45 [2.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.35 [2.24]	0.43 [2.76]	0.51 [3.88]	0.49 [3.66]	0.40 [3.04]	0.39 [2.83]
Quintile	NYSE	EW	0.38 [2.01]	0.44 [2.28]	0.35 [2.06]	0.28 [1.65]		
Quintile	Name	VW	0.60 [2.61]	0.91 [4.23]	0.82 [4.56]	0.75 [4.20]	0.36 [2.37]	0.31 [2.07]
Quintile	Cap	VW	0.30 [2.15]	0.37 [2.57]	0.44 [3.62]	0.41 [3.35]	0.36 [2.94]	0.33 [2.68]
Decile	NYSE	VW	0.56 [2.72]	0.77 [3.75]	0.78 [4.22]	0.74 [4.04]	0.50 [2.84]	0.48 [2.72]

Table 3: Conditional sort on size and DSC

This table presents results for conditional double sorts on size and DSC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DSC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DSC and short stocks with low DSC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DSC Quintiles					DSC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.26 [0.58]	0.26 [0.75]	0.56 [1.92]	1.05 [3.50]	1.03 [3.59]	0.77 [2.89]	1.02 [3.91]	0.82 [3.96]	0.79 [3.71]	0.38 [2.10]	0.35 [1.94]
	(2)	0.16 [0.39]	0.69 [2.48]	0.92 [3.31]	0.76 [2.80]	0.88 [3.11]	0.72 [3.07]	0.97 [4.29]	0.82 [4.25]	0.77 [3.94]	0.28 [1.71]	0.26 [1.57]
	(3)	0.23 [0.70]	0.87 [3.52]	1.00 [3.82]	0.73 [2.76]	0.82 [3.01]	0.59 [3.37]	0.76 [4.43]	0.71 [4.22]	0.64 [3.78]	0.31 [1.98]	0.27 [1.74]
	(4)	0.55 [1.89]	0.81 [3.66]	0.86 [3.64]	0.80 [3.20]	0.81 [3.18]	0.27 [1.82]	0.36 [2.50]	0.34 [2.36]	0.29 [2.01]	-0.01 [-0.04]	-0.03 [-0.22]
	(5)	0.57 [2.21]	0.63 [3.40]	0.88 [4.60]	0.71 [3.75]	0.82 [3.20]	0.25 [1.47]	0.25 [1.44]	0.38 [2.80]	0.36 [2.57]	0.33 [2.33]	0.31 [2.18]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DSC Quintiles					DSC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	337	335	342	346	347	32	28	34	43	52	
	(2)	101	101	101	101	101	66	71	73	71	73	
	(3)	71	71	72	71	72	123	130	129	128	130	
	(4)	62	62	62	62	62	276	298	288	281	285	
(5)	59	59	58	59	59	2197	2210	3043	2576	2040		

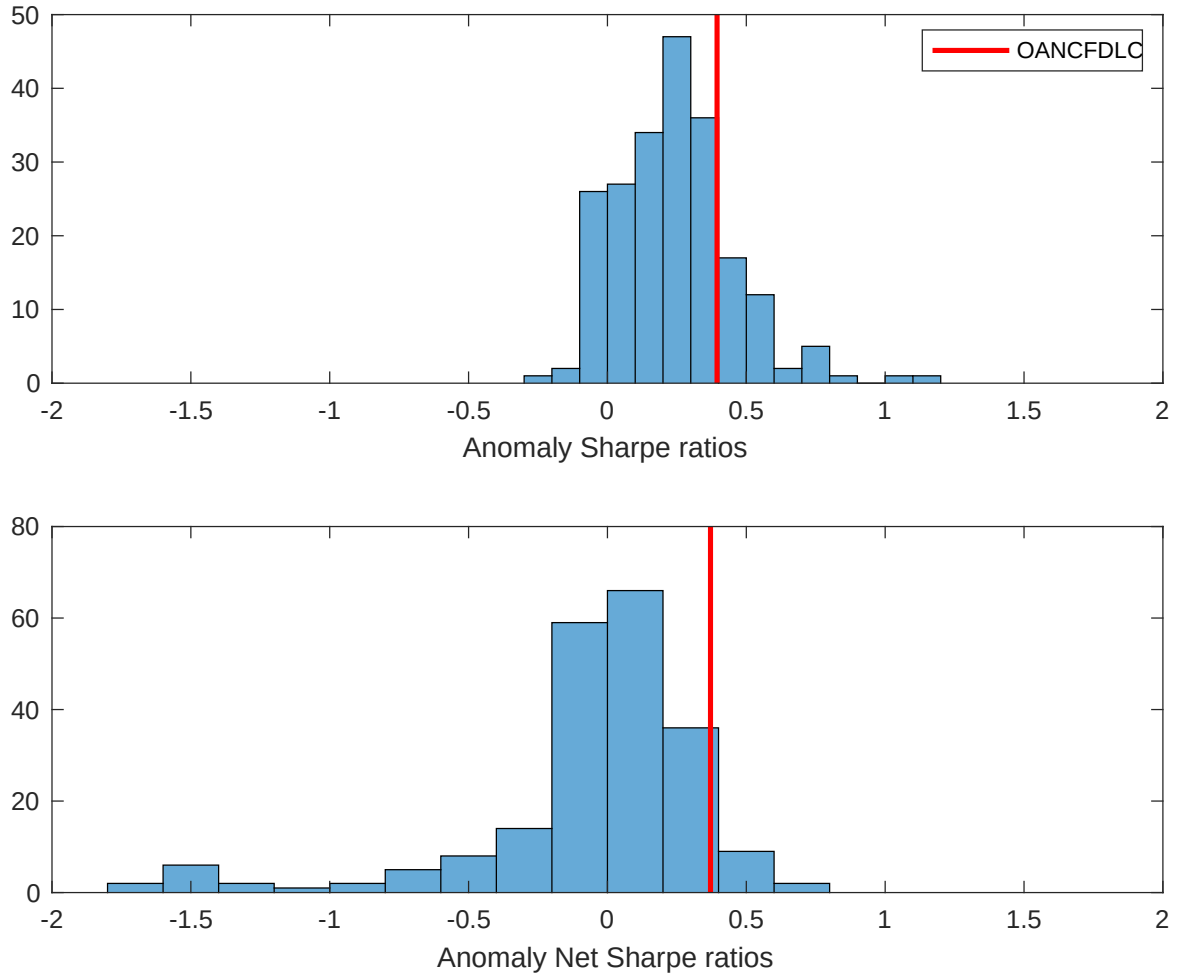


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DSC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

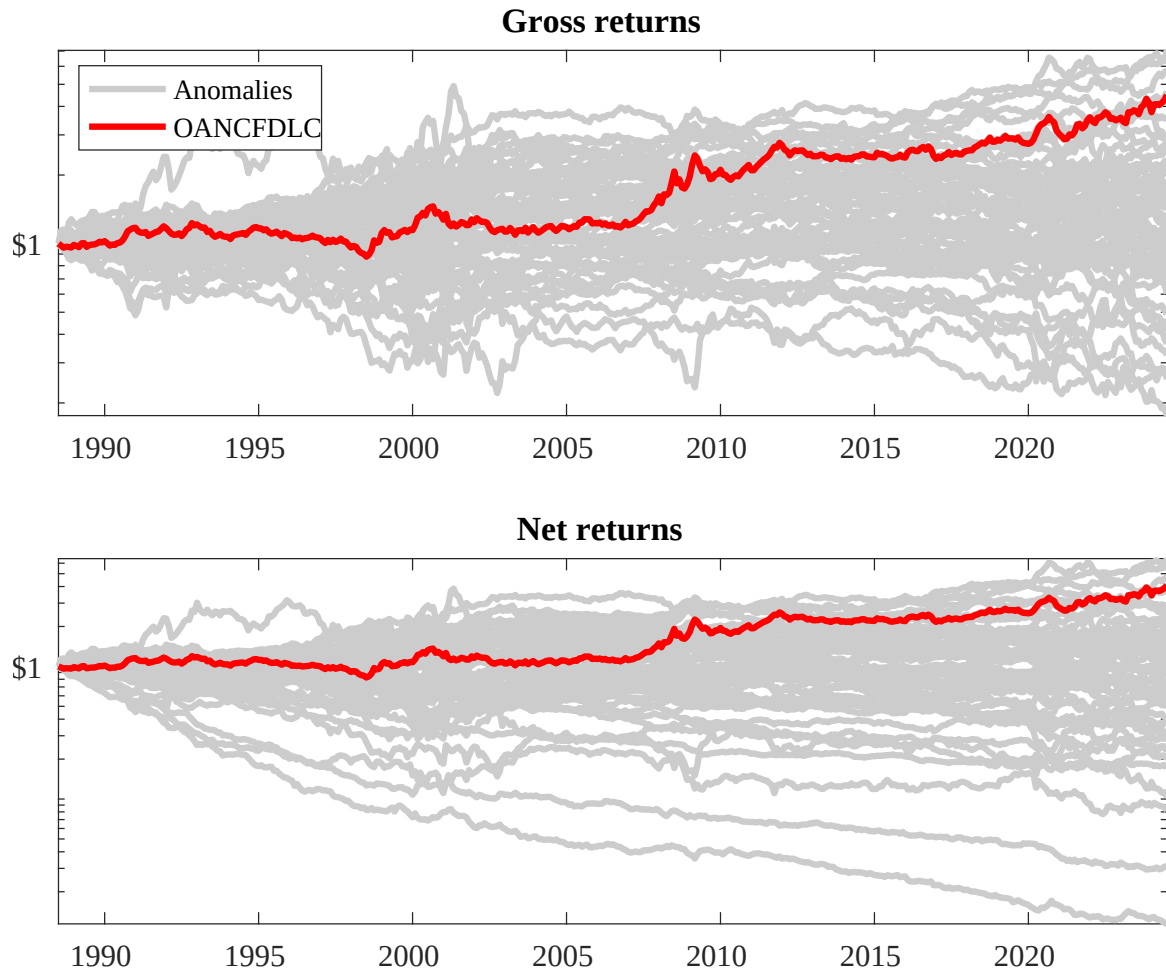
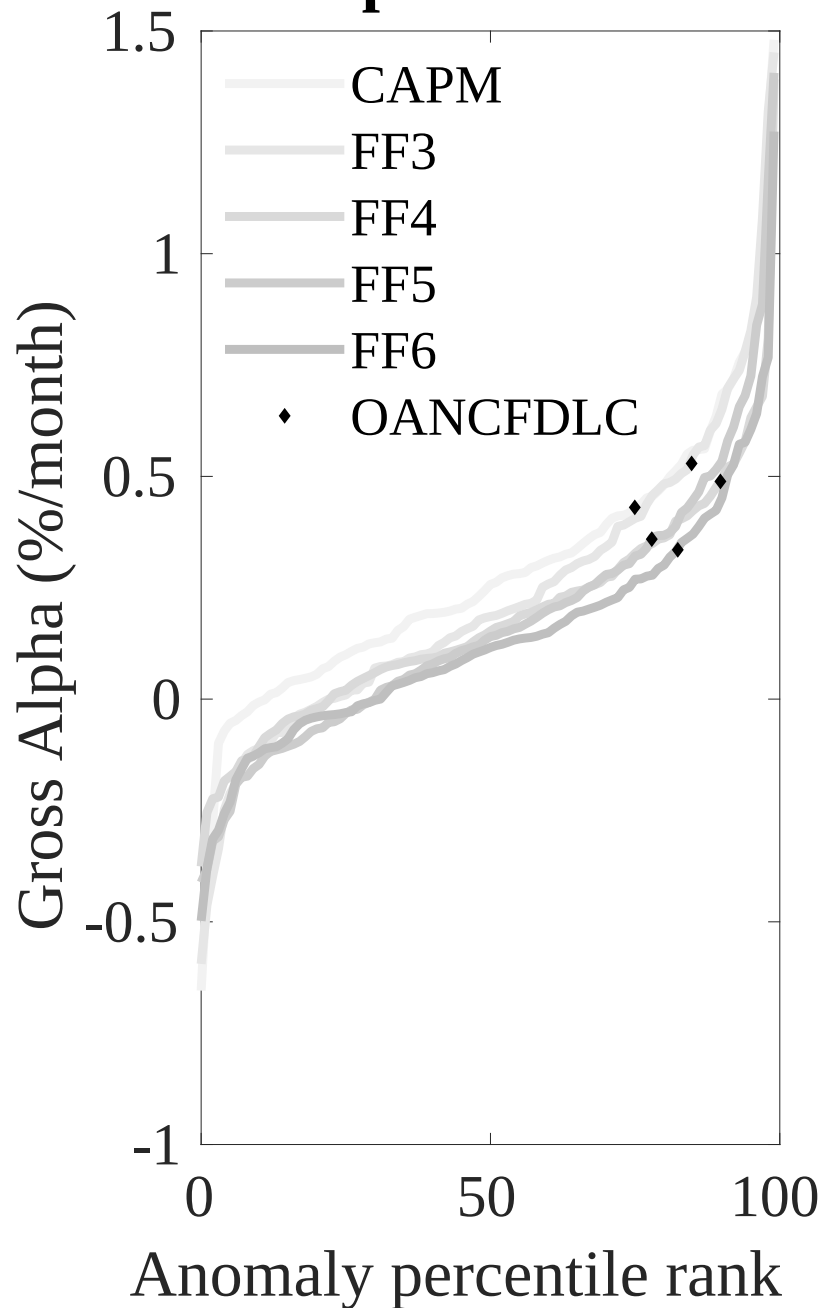


Figure 3: Dollar invested.

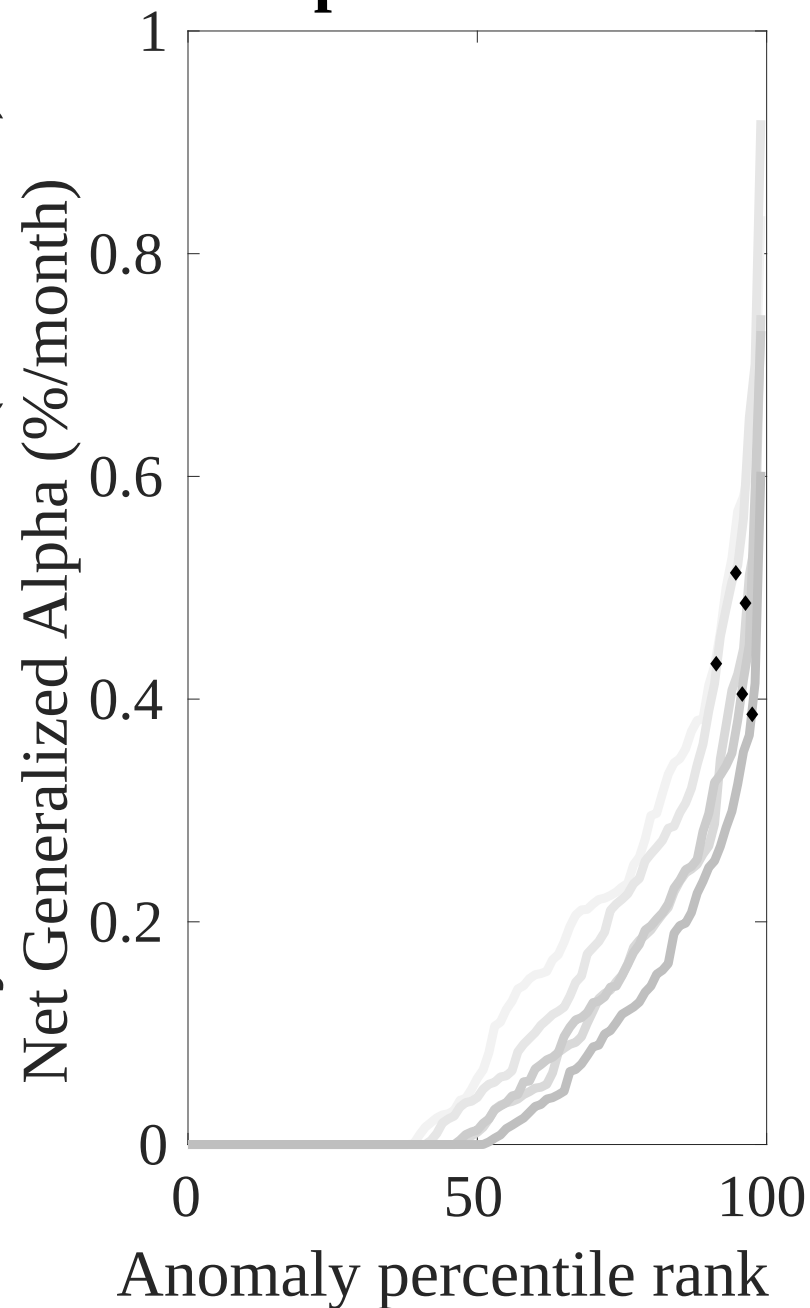
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DSC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



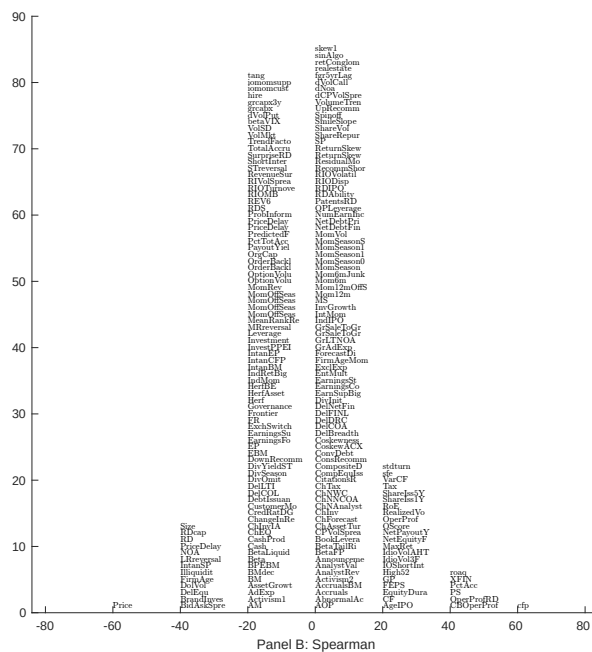
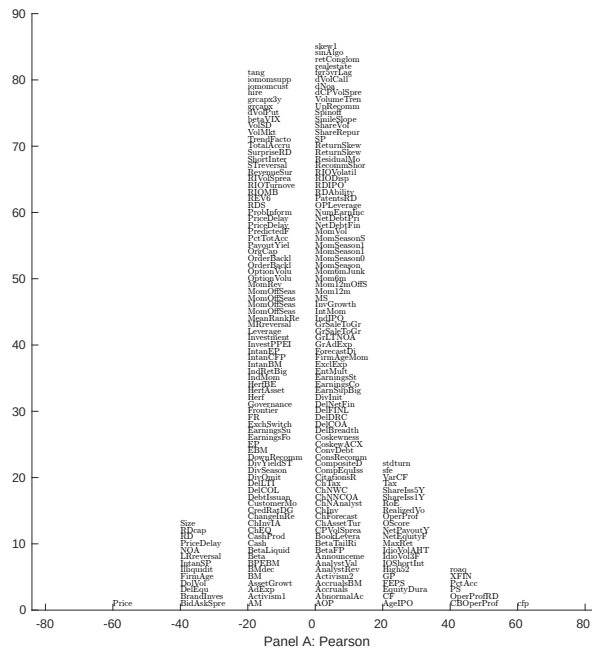
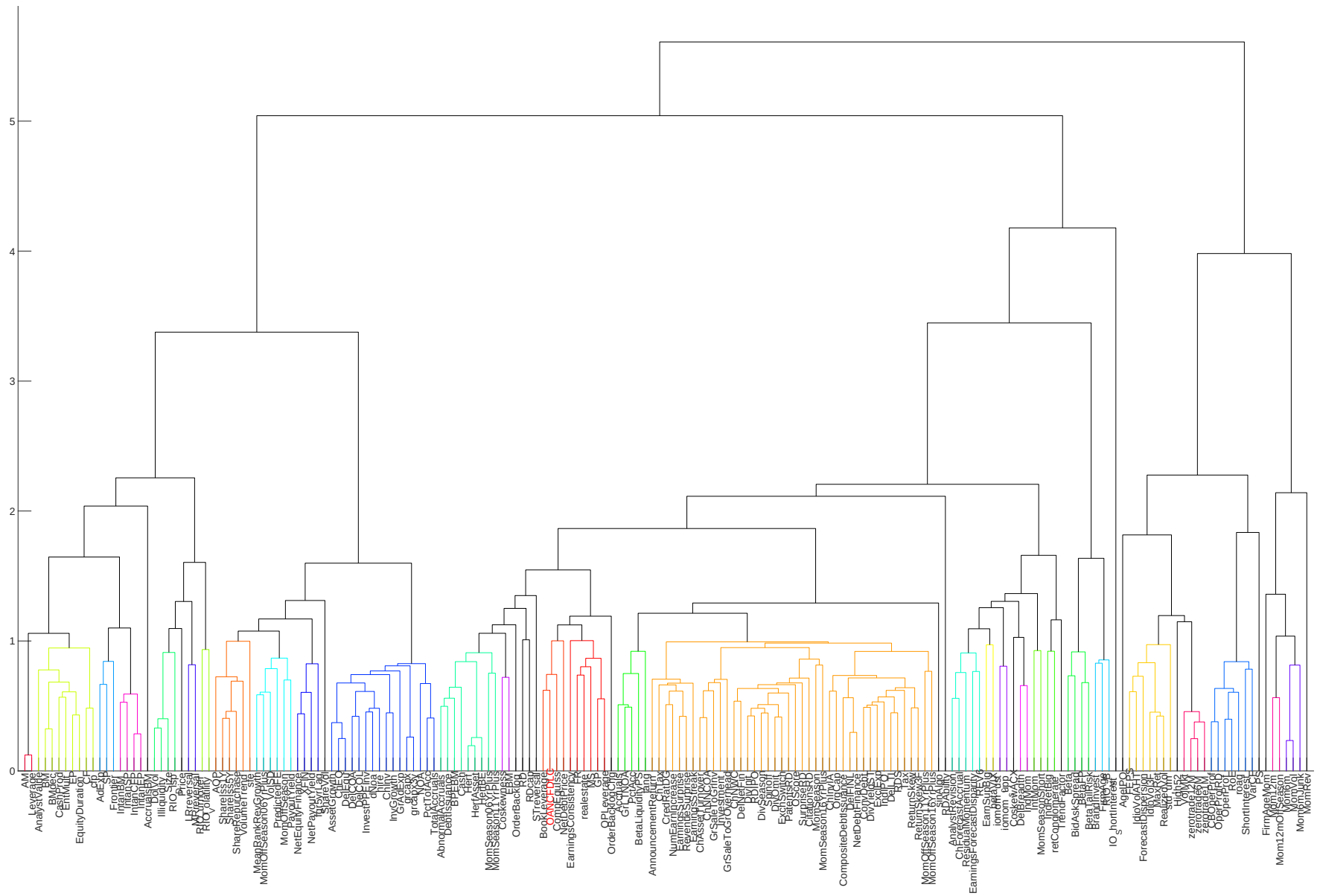


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with DSC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



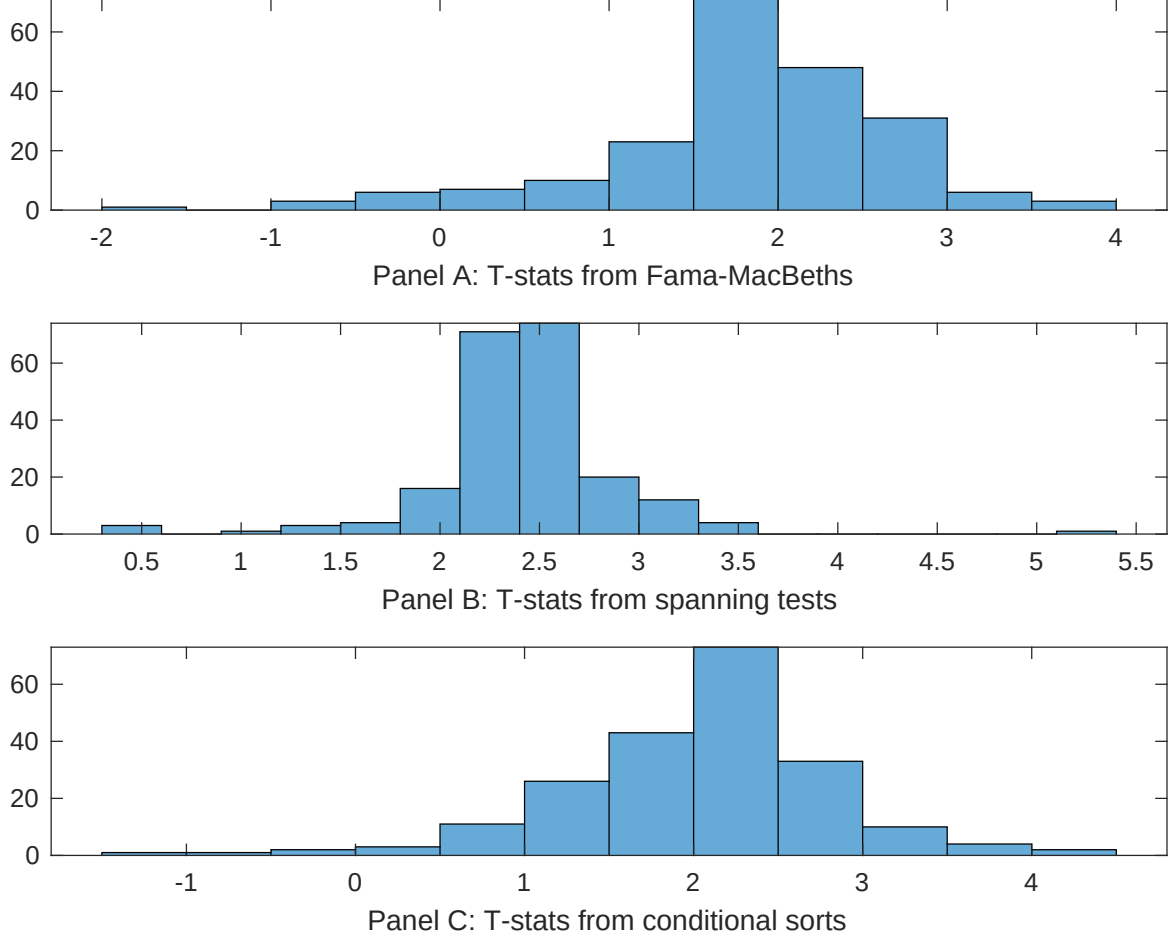


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DSC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSC} DSC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DSC. Stocks are finally grouped into five DSC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DSC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DSC}DSC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.90 [2.83]	0.95 [3.26]	0.11 [3.96]	0.14 [4.58]	0.85 [2.54]	1.00 [3.61]	0.15 [4.85]
DSC	0.24 [0.05]	0.71 [1.42]	-0.18 [-0.49]	0.13 [2.04]	0.20 [0.43]	0.10 [1.53]	-0.46 [-1.35]
Anomaly 1	0.21 [4.70]						-0.70 [-1.00]
Anomaly 2		0.10 [3.88]					0.46 [1.63]
Anomaly 3			0.49 [2.84]				0.45 [2.81]
Anomaly 4				0.53 [3.76]			0.87 [7.79]
Anomaly 5					0.20 [3.76]		0.18 [2.63]
Anomaly 6						0.13 [3.09]	0.17 [2.79]
# months	427	427	427	432	427	432	427
$\bar{R}^2(\%)$	1	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DSC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DSC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.25 [1.86]	0.32 [2.33]	0.30 [2.25]	0.59 [5.10]	0.29 [2.12]	0.24 [2.08]	0.32 [3.19]
Anomaly 1	27.03 [4.61]						31.01 [4.80]
Anomaly 2		11.55 [2.17]					4.32 [1.10]
Anomaly 3			26.44 [4.47]				15.54 [3.33]
Anomaly 4				-61.30 [-12.59]			-49.69 [-10.80]
Anomaly 5					12.76 [2.28]		-4.57 [-0.75]
Anomaly 6						57.51 [13.39]	37.07 [9.10]
mkt	-2.31 [-0.69]	-4.71 [-1.40]	-0.79 [-0.23]	-1.71 [-0.59]	-2.51 [-0.72]	-3.43 [-1.21]	2.37 [0.93]
smb	7.60 [1.53]	-0.66 [-0.14]	6.37 [1.30]	-15.44 [-3.57]	4.49 [0.89]	8.70 [2.12]	2.02 [0.51]
hml	-55.57 [-8.93]	-74.09 [-10.66]	-57.44 [-9.40]	-44.90 [-8.45]	-60.90 [-9.62]	-45.27 [-8.72]	-23.10 [-4.06]
rmw	12.78 [1.91]	22.58 [3.55]	2.39 [0.30]	21.79 [4.17]	16.97 [2.30]	40.31 [7.72]	3.66 [0.56]
cma	13.42 [1.58]	14.79 [1.69]	17.09 [2.03]	23.71 [3.23]	17.93 [2.09]	-3.24 [-0.44]	1.36 [0.21]
umd	2.03 [0.68]	9.50 [2.48]	-2.48 [-0.75]	8.11 [3.15]	2.63 [0.85]	0.98 [0.39]	1.25 [0.42]
# months	428	428	428	432	428	432	428
$\bar{R}^2(\%)$	36	33	36	51	33	53	66

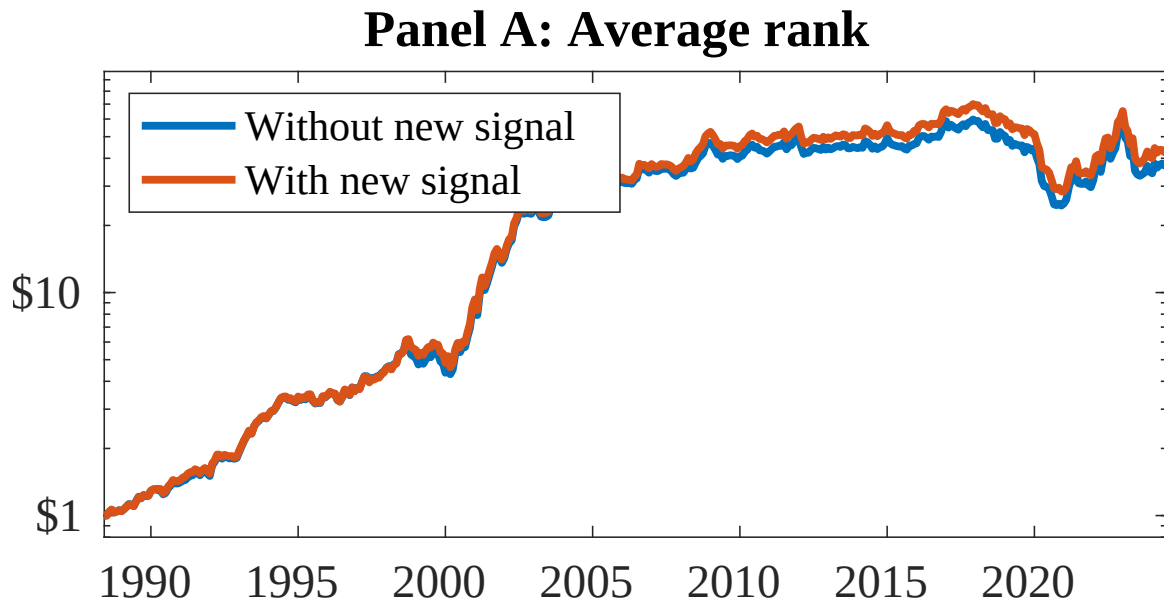


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as DSC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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